

Math 3IA3 - Assignment 4

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Exercise 4.2.12

Find functions f and g defined on $(0, \infty)$ such that $\lim_{x \rightarrow \infty} f = \infty$ and $\lim_{x \rightarrow \infty} g = \infty$ and $\lim_{x \rightarrow \infty} (f - g) = 0$. Can you find such functions, with $g(x) > 0$ for all $x > 0$ for all $x \in (0, \infty)$, such that $\lim_{x \rightarrow \infty} \frac{f}{g} = 0$?

Solution:

(I) Find f and g such that $\lim_{x \rightarrow \infty} (f - g) = 0$ while both diverge.

A suitable choice is

$$f(x) = x + \frac{1}{x}, \quad g(x) = x.$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(x + \frac{1}{x} \right) = \infty,$$

$$\lim_{x \rightarrow \infty} g(x) = \lim_{x \rightarrow \infty} x = \infty,$$

$$\lim_{x \rightarrow \infty} (f(x) - g(x)) = \lim_{x \rightarrow \infty} \frac{1}{x} = 0.$$

(II) Find f and g with $g(x) > 0$ and $\lim_{x \rightarrow \infty} \frac{f}{g} = 0$.

Let

$$f(x) = \sqrt{x}, \quad g(x) = x.$$

$$\lim_{x \rightarrow \infty} f(x) = \infty,$$

$$\lim_{x \rightarrow \infty} g(x) = \infty, \quad g(x) > 0 \quad \forall x > 0,$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{x} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{x}} = 0.$$

Exercise 5.1.12

Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}$ and that $f(r) = 0$ for every rational number r . Prove that $f(x) = 0$ for all $x \in \mathbb{R}$.

Solution:

Assume, for contradiction, that there exists $k \in \mathbb{R}$ with $f(k) \neq 0$. Let $L = f(k)$, so $|L| > 0$. Since f is continuous at k , for

$$\varepsilon = \frac{|L|}{2} > 0,$$

there exists $\delta > 0$ such that

$$0 < |x - k| < \delta \implies |f(x) - L| < \varepsilon.$$

Because $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists a rational number r with

$$0 < |r - k| < \delta.$$

Then

$$|f(r) - L| = |0 - L| = |L| \geq \frac{|L|}{2} = \varepsilon,$$

contradicting continuity at k .

Thus no such k exists and $f(x) = 0$ for all real x .

Exercise 5.3.12

Let $I := [0, \frac{\pi}{2}]$ and let $f: I \rightarrow \mathbb{R}$ be defined by $f(x) := \sup \{x^2, \cos(x)\}$ for $x \in I$. Show that there exists an absolute maximum point $x_0 \in I$ for f on I . Show that x_0 is a solution to the equation $\cos(x) = x^2$.

Solution:

(I) Show that f attains an absolute maximum on I .

Both x^2 and $\cos x$ are continuous on I . Thus

$$f(x) = \max\{x^2, \cos x\}$$

is continuous as the maximum of two continuous functions. Since I is compact, f attains an absolute maximum at some $x_0 \in I$.

(II) Show that any interior maximizer satisfies $x_0^2 = \cos x_0$.

Suppose $x_0 \in (0, \frac{\pi}{2})$ is a point where f achieves its maximum.

Case 1: $\cos(x_0) > x_0^2$. Then in a small neighbourhood of x_0 we have $f(x) = \cos x$. Since $\cos x$ is strictly decreasing on $[0, \frac{\pi}{2}]$, choosing $x < x_0$ sufficiently close gives $\cos(x) > \cos(x_0)$, contradicting maximality.

Case 2: $x_0^2 > \cos(x_0)$. Then near x_0 we have $f(x) = x^2$. Since x^2 is strictly increasing on $[0, \frac{\pi}{2}]$, choosing $x > x_0$ sufficiently close gives $x^2 > x_0^2$, again contradicting maximality.

Thus neither case is possible, so

$$x_0^2 = \cos(x_0).$$

Exercise 5.6.7

Let $I \subseteq \mathbb{R}$ be an interval and let $f: I \rightarrow \mathbb{R}$ be increasing on I . If c is not an endpoint of I , show that the jump $j_f(c)$ of f at c is given as

$$\inf\{f(y) - f(x) : x < c < y, x, y \in I\}.$$

Solution:

Let

$$j_f(c) = f(c+) - f(c-),$$

where $f(c+) = \lim_{y \downarrow c} f(y)$ and $f(c-) = \lim_{x \uparrow c} f(x)$ exist because f is increasing.

Let

$$S = \{f(y) - f(x) : x < c < y, x, y \in I\}.$$

(I) Show $j_f(c)$ is a lower bound of S .

If $x < c < y$, then by monotonicity,

$$f(y) - f(x) \geq f(c+) - f(c-) = j_f(c).$$

(II) Show $\inf S \leq j_f(c)$.

Let $\varepsilon > 0$. Choose $y > c$ close enough so that

$$f(y) < f(c+) + \varepsilon/2,$$

and choose $x < c$ close enough so that

$$f(x) > f(c-) - \varepsilon/2.$$

Then

$$\begin{aligned} f(y) - f(x) &< (f(c+) + \varepsilon/2) - (f(c-) - \varepsilon/2) \\ &= j_f(c) + \varepsilon. \end{aligned}$$

Thus $\inf S \leq j_f(c) + \varepsilon$ for all $\varepsilon > 0$, so $\inf S \leq j_f(c)$.

Combining both steps gives $\inf S = j_f(c)$.

Exercise 6.1.9

Show that if f is an even function and has a derivative at every point, then the derivative f' is an odd function. Also prove that if $g: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable and odd function then g' is an even function.

Solution:

(I) If f is even and differentiable, then f' is odd.

Since f is even, $f(-x) = f(x)$ for all x .

For any x ,

$$\begin{aligned} f'(-x) &= \lim_{h \rightarrow 0} \frac{f(-x+h) - f(-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x-h) - f(x)}{h} \\ &= \lim_{k \rightarrow 0} \frac{f(x+k) - f(x)}{-k} \quad (k = -h) \\ &= -f'(x). \end{aligned}$$

Thus f' is odd.

(II) If g is odd and differentiable, then g' is even.

Since g is odd, $g(-x) = -g(x)$.

For any x ,

$$\begin{aligned} g'(-x) &= \lim_{h \rightarrow 0} \frac{g(-x+h) - g(-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-g(x-h) + g(x)}{h} \\ &= \lim_{k \rightarrow 0} \frac{g(x+k) - g(x)}{k} \quad (k = -h) \\ &= g'(x). \end{aligned}$$

Thus g' is even.